Linear Algebra and Its Applications

Section 8.3: Convex Combinations

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

8.3 Convex combinations 3

Practice problem 8.3.P1: Test a triangle in three-dimensional space 3

Practice problem 8.3.P2: The hull of the positive reciprocal curve 3

Exercise 8.3.1: A triangle with a missing upper edge 3

Exercise 8.3.2: Three convex hulls of graphs 3

Exercise 8.3.3: Use the unique coordinates from Section 8.1, Exercise 5 4

Exercise 8.3.4: A second list with the same orthogonal generators 4

Exercise 8.3.5: Convex coordinates in a tetrahedron 5

Exercise 8.3.6: Distinguish the linear, affine, and convex hulls 5

Exercise 8.3.7: Barycentric coordinates locate four planar points 5

Exercise 8.3.8: Another barycentric-coordinate table 6

Exercise 8.3.9: Read positions from tetrahedral coordinates 6

Exercise 8.3.10: A second tetrahedral list 6
Exercise 8.3.11: Three statements about convexity 6
Exercise 8.3.12: Three more statements about hulls 6
Exercise 8.3.13: A linear image preserves convexity 7
Exercise 8.3.14: A linear preimage preserves convexity 7
Exercise 8.3.15: Eliminate a vertex in two different ways 7
Exercise 8.3.16: A second coefficient-elimination example 7
Exercise 8.3.17: A containing convex set contains the hull 7
Exercise 8.3.18: Convex hull is monotone under inclusion 8
Exercise 8.3.19: Union of hulls need not fill the combined hull 8
Exercise 8.3.20: Intersection of hulls can be larger 8
Exercise 8.3.21: The midpoint construction for a quadratic curve 8
Exercise 8.3.22: A three-quarter construction 9
Exercise 8.3.23: Expand the quadratic Bézier formula 9
Exercise 8.3.24: The cubic curve also stays inside its controls 9

8.3 Convex combinations

A linear combination allows arbitrary coefficients. An affine combination requires that their sum be 1. A convex combination requires both sum 1 and *nonnegative* coefficients. Thus $\operatorname{conv} S \subseteq \operatorname{aff} S \subseteq \operatorname{Span} S$. For an affinely independent list, its affine coordinates are unique: a negative coordinate rules out convex membership, all positive coordinates put the point in the relative interior of its simplex, and zero coordinates locate it on a face. A convex hull consists of finite convex combinations; it need not include limit points not obtained by those combinations.

Practice problem 8.3.P1: Test a triangle in three-dimensional space

Use the supplied vertices $v_1 = (6, 2, 2)$, $v_2 = (7, 1, 5)$, $v_3 = (-2, 4, -1)$. To find affine coordinates solve $p - v_1 = a(v_2 - v_1) + b(v_3 - v_1)$, then use coefficients $(1 - a - b, a, b)$. For $p_1 = (1, 3, 1)$ the solution is $a = 1/3, b = 2/3$; checking gives $p_1 = 0v_1 + (1/3)v_2 + (2/3)v_3$. All coefficients are nonnegative, so $p_1 \in \operatorname{conv} S$. For $p_2 = (3, 2, 1)$ the first two coordinate equations give $a = 1, b = 1/2$, but the third would require $3a - 3b = -1$, whereas it equals $3/2$. Thus p_2 is not even in the affine hull and cannot be in the convex hull.

Practice problem 8.3.P2: The hull of the positive reciprocal curve

What we must show. Describe the hull of $S = \{(x, 1/x) : x > 0\}$ exactly. No convex combination has $x \le 0$. Moreover convexity of $1/x$ on positive inputs gives $\sum_i \lambda_i/x_i \ge 1/(\sum_i \lambda_i x_i)$. For example this follows directly from Cauchy–Schwarz applied to $\sqrt{\lambda_i x_i}$ and $\sqrt{\lambda_i/x_i}$. Hence the hull lies in $\{(x, y) : x > 0, y \ge 1/x\}$. Conversely, for a point (x, y) in that set draw the line of slope -1 through it, $u + v = x + y$. Its intersections with $uv = 1$ have positive x-coordinates given by roots of $u^2 - (x + y)u + 1 = 0$. The polynomial at $u = x$ is $1 - xy \le 0$, so x lies between the two roots. Thus (x, y) lies on the line segment joining these two curve points, or is itself the single tangency point. It is a convex combination of them. Therefore the hull is exactly the region on and above the curve.

Exercise 8.3.1: A triangle with a missing upper edge

Every convex combination of vertical-segment points and $(2, 0)$ can be written $(1 - \lambda)(0, a) + \lambda(2, 0)$ with $0 \le \lambda \le 1$ and $0 \le a < 1$. (Combine all the vertical points into their weighted average first.) Thus the hull is

$$\{(x, y) : 0 \le x < 2,\ 0 \le y < 1 - x/2\} \ \cup\ \{(2, 0)\}.$$

Conversely any point in the first set uses $\lambda = x/2$ and $a = y/(1 - x/2)$, which satisfy the required inequalities. The lower edge and right vertex are included, but the sloping upper edge and the point $(0, 1)$ are excluded. Replacing the strict inequality by a weak one would add points not actually in the convex hull.

Exercise 8.3.2: Three convex hulls of graphs

(a) The reciprocal graph starting at $x = 1/2$.

The exact hull is

$$\{(1/2, 2)\} \ \cup\ \{(x, y) : x > 1/2,\ 1/x \le y < 2\}.$$

Every combination has $x \geq 1/2$, lies on or above the reciprocal curve by convexity, and has $y \leq 2$. Equality $y = 2$ forces every positive-weight input to be the endpoint $(1/2, 2)$, so it occurs only there. For the converse fix $x > 1/2$ and $1/x \leq y < 2$. Put $b = (2x-1)/(2-y)$. The lower bound on y gives $b \geq x$. The chord joining $(1/2, 2)$ and $(b, 1/b)$ has height $2 - (2x-1)/b = y$ at horizontal coordinate x. Because $1/2 < x \leq b$, the point lies on that segment and is in the hull.

(b) The sine graph on the whole real line.

The hull is the closed strip $\{(x, y) : -1 \leq y \leq 1\}$. The range of sine bounds every combination within the strip. Conversely its maximum points $(\pi/2 + 2j\pi, 1)$ extend in both horizontal directions. For any fixed real x, two such points bracket x, so their segment contains $(x, 1)$. The minimum points similarly give $(x, -1)$. The segment between these two hull points contains every (x, y) in the strip.

(c) The square-root graph.

The hull is

$$\{(0,0)\} \ \cup \ \{(x,y) : x > 0, \ 0 < y \leq \sqrt{x}\}.$$

Concavity of square root bounds the height of a convex combination by $\sqrt{x}$; nonnegativity is automatic. If $x > 0$, some positive-weight input has positive x and therefore positive height, so height cannot be zero. Conversely, for $x > 0$ and $0 < y \leq \sqrt{x}$, take $a = (x/y)^2$ and $\lambda = y^2/x \in (0, 1]$. Then $\lambda(a, \sqrt{a}) + (1-\lambda)(0,0) = (x, y)$. This proves the formula and explains why the positive horizontal axis is not included.

Exercise 8.3.3: Use the unique coordinates from Section 8.1, Exercise 5

The three orthogonal generators are $b_1 = (2, 1, 1)$, $b_2 = (1, 0, -2)$, $b_3 = (2, -5, 1)$. Their squared lengths are $6, 5, 30$, so the unique linear coefficients of p are $c_j = (p \cdot b_j)/(b_j \cdot b_j)$. For the three supplied points the results are

p	(c_1, c_2, c_3)	sum
$(3, 8, 4)$	$(3, -1, -1)$	1
$(6, -3, 3)$	$(2, 0, 1)$	3
$(0, -1, -5)$	$(-1, 2, 0)$	1

The first and third have negative coordinates; the second does not even have coefficient sum one. Since the linear representation in a basis is unique, no alternative nonnegative sum-one representation exists.

Conclusion. None of the three points is in $\operatorname{conv} S$.

Exercise 8.3.4: A second list with the same orthogonal generators

The same dot-product calculation gives

p	(c_1, c_2, c_3)	sum
$(0, -19, -5)$	$(-4, 2, 3)$	1
$(3/2, -13/10, -1/2)$	$(1/5, 1/2, 3/10)$	1
$(5, -4, 0)$	$(1, 1, 1)$	3

Only the middle row has all coefficients nonnegative and sum one. Thus $p_2 \in \operatorname{conv} S$, while p_1, p_3 are not.

Exercise 8.3.5: Convex coordinates in a tetrahedron

Append a final coordinate 1 to each of the four supplied v-vectors. Solving the resulting 4×4 systems for the two points gives

$$[p_1]_S = (-1/6, 1/3, 2/3, 1/6), \qquad [p_2]_S = (1/3, 1/3, 1/6, 1/6).$$

Here coordinates refer to affine/barycentric coordinates. The appended matrix is invertible, so these representations are unique. For instance $(1/3)v_1 + (1/3)v_2 + (1/6)v_3 + (1/6)v_4 = (0, -2, 2) = p_2$. The first representation has a negative coefficient and therefore is not convex; the second has all coefficients positive and sum one.

Conclusion. p_1 is outside the convex hull; p_2 is in it.

Exercise 8.3.6: Distinguish the linear, affine, and convex hulls

The supplied orthogonal vectors are $v_1 = (2, 0, -1, 2)$, $v_2 = (0, -2, 2, 1)$, $v_3 = (-2, 1, 0, 2)$; each has squared length 9. For a proposed p, first compute $c_j = (p \cdot v_j)/9$ and check whether $\sum c_j v_j = p$. Only after this check do sum-one and sign tests apply.

(a) For $p_1 = (-1, 2, -3/2, 5/2)$, the coefficients are $(1/2, -1/2, 1)$ and reconstruct p_1. They sum to one but include a negative coefficient. Thus p_1 is in the span and affine hull, but not the convex hull.

(b) For $p_2 = (-1/2, 0, 1/4, 7/4)$, the coefficients are $(1/4, 1/4, 1/2)$. They reconstruct p_2, sum to one, and are nonnegative. Thus p_2 is in all three hulls.

(c) For $p_3 = (6, -4, 1, -1)$, the coefficients are $(1, 1, -2)$ and reconstruct p_3, but their sum is zero. Hence p_3 is in the span only, not the affine or convex hull.

(d) For $p_4 = (-1, -2, 0, 4)$, the dot-product coefficients are $(2/3, 8/9, 8/9)$, but the reconstruction error is $p_4 - \sum c_j v_j = (-5/9, -10/9, -10/9, 0) \neq 0$. Thus p_4 is not in the span and consequently in neither smaller hull.

Exercise 8.3.7: Barycentric coordinates locate four planar points

(a) Solve the affine-coordinate systems.

With $T = \{(-1, 0), (2, 3), (4, 1)\}$, solve

$$\begin{pmatrix} -1 & 2 & 4 \\ 0 & 3 & 1 \\ 1 & 1 & 1 \end{pmatrix} c = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}.$$

The four answers, in the listed order, are

$$(1/3, 1/6, 1/2), \quad (0, 1/2, 1/2), \quad (1/2, -1/4, 3/4), \quad (1/2, 3/4, -1/4).$$

(b) Use signs and zero entries.

p_1 has all positive entries and is inside the triangle. p_2 has one zero and the other entries positive, so it is on the edge joining vertices two and three. p_3, p_4 have negative entries and are outside.

Exercise 8.3.8: Another barycentric-coordinate table

For $T = \{(2,0),(0,5),(-1,1)\}$, use the appended-coordinate system

$$\begin{pmatrix} 2 & 0 & -1 \\ 0 & 5 & 1 \\ 1 & 1 & 1 \end{pmatrix} c = (x, y, 1)^T.$$

For p_1, p_2, p_3, p_4 its solutions are respectively

$$(12/13, 3/13, -2/13), \quad (8/13, 2/13, 3/13), \quad (2/3, 0, 1/3), \quad (9/13, -1/13, 5/13).$$

Thus p_1 is outside, p_2 inside, p_3 on the edge joining vertices one and three, and p_4 outside. The p_3 supplied in the exercise is $(1, 1/3)$, not $(1, 1)$; preserving this fraction is essential.

Exercise 8.3.9: Read positions from tetrahedral coordinates

A point is inside precisely when all four barycentric coordinates are positive. Nonnegative coordinates with a zero put it on a surface; two zeros put it on an edge. A negative coordinate puts it outside. Therefore $p_1 = (2, 0, 0, -1)$ is outside; $p_2 = (0, 1/2, 1/4, 1/4)$ is on the face opposite vertex one, but not an edge; $p_3 = (1/2, 0, 3/2, -1)$ is outside; $p_4 = (1/3, 1/4, 1/4, 1/6)$ is inside; and $p_5 = (1/3, 0, 2/3, 0)$ is on the edge joining vertices one and three.

Exercise 8.3.10: A second tetrahedral list

Using the same criteria, $q_1 = (1/8, 1/4, 1/8, 1/2)$ is inside; $q_2 = (3/4, -1/4, 0, 1/2)$ is outside; $q_3 = (0, 3/4, 1/4, 0)$ is on the edge joining vertices two and three; $q_4 = (0, -2, 0, 3)$ is outside; and $q_5 = (1/3, 1/3, 1/3, 0)$ is on the face opposite vertex four, not an edge.

Exercise 8.3.11: Three statements about convexity

(a) **False.** Sum one makes a combination affine, but negative coefficients are still possible. Convexity also requires each coefficient nonnegative.

(b) **False.** A singleton $S = \{v\}$ has $\operatorname{conv} S = S$. More generally this equality holds for any convex S.

(c) **False.** Two singleton sets are convex but their union $\{(0,0),(2,0)\}$ misses the midpoint $(1,0)$.

Exercise 8.3.12: Three more statements about hulls

(a) **True.** Containing every segment between two of its points is exactly the definition of convexity.

(b) **True.** If two points belong to both convex sets, their whole joining segment belongs to both, hence to the intersection. The empty intersection is also convex under the usual vacuous definition.

(c) **False.** Carathéodory's theorem says *at most* six points in $\mathbb{R}^5$, not necessarily six distinct available points. A singleton S is an immediate counterexample to the stated requirement.

Exercise 8.3.13: A linear image preserves convexity

For $u = f(x), v = f(y)$ in $f(S)$ and $0 \le t \le 1$, linearity gives $(1-t)u + tv = f((1-t)x + ty)$. Convexity of S puts the input $(1-t)x + ty$ in S, so this output is in $f(S)$. Hence $f(S)$ is convex. Empty sets cause no exception.

Exercise 8.3.14: A linear preimage preserves convexity

For $x, y \in S$, the images $f(x), f(y)$ lie in T. Their combination $(1-t)f(x) + tf(y)$ lies in T for $0 \le t \le 1$ by convexity. This equals $f((1-t)x + ty)$ by linearity, so the input $(1-t)x + ty$ lies in S. Thus the whole segment is in S.

Exercise 8.3.15: Eliminate a vertex in two different ways

Direct coordinate addition checks both supplied identities: $p = (1/3)v_1 + (1/3)v_2 + (1/6)v_3 + (1/6)v_4 = (2, 1)$ and $v_1 - v_2 + v_3 - v_4 = 0$. Add s times the zero relation to the coefficients. They become

$$(1/3 + s,\ 1/3 - s,\ 1/6 + s,\ 1/6 - s).$$

Their sum stays one and their combination stays p. For $s = 1/6$ and $s = -1/6$ all entries are nonnegative and one is zero, giving respectively

$$p = \tfrac{1}{2}v_1 + \tfrac{1}{6}v_2 + \tfrac{1}{3}v_3, \qquad p = \tfrac{1}{6}v_1 + \tfrac{1}{2}v_2 + \tfrac{1}{3}v_4.$$

This is the coefficient-elimination step in the proof of Carathéodory's theorem.

Exercise 8.3.16: A second coefficient-elimination example

The printed instruction says "Repeat Exercise 9," but the supplied convex combination and zero relation clearly call for the procedure of Exercise 15. We carry out that intended task explicitly. The given coefficients are $(1, 72, 37, 11)/121$, whose sum is one; coordinate substitution gives $p = (1, 2)$. The relation $10v_1 - 6v_2 + 7v_3 - 11v_4 = 0$ also has coefficient sum zero. Add $s(10, -6, 7, -11)$ to the original coefficients. For $s = 1/121$, the last coefficient becomes zero and we obtain

$$p = \tfrac{1}{11}v_1 + \tfrac{6}{11}v_2 + \tfrac{4}{11}v_3.$$

For $s = -1/1210$, the first becomes zero and the result is

$$p = \tfrac{3}{5}v_2 + \tfrac{3}{10}v_3 + \tfrac{1}{10}v_4.$$

Each list is nonnegative, sums to one, and reconstructs p.

Exercise 8.3.17: A containing convex set contains the hull

If $A \subseteq B$ and B is convex, then any finite convex combination of points in A is a convex combination of points in B and therefore belongs to B. Every point of $\operatorname{conv} A$ is such a combination, so $\operatorname{conv} A \subseteq B$.

Exercise 8.3.18: Convex hull is monotone under inclusion

If $A \subseteq B$, every convex combination of points in A is also one of points in B. Therefore $\operatorname{conv} A \subseteq \operatorname{conv} B$. This also follows from the preceding exercise because $\operatorname{conv} B$ is convex and contains A.

Exercise 8.3.19: Union of hulls need not fill the combined hull

(a) Prove the inclusion.

Since $A \subseteq A \cup B$ and $B \subseteq A \cup B$, monotonicity gives both $\operatorname{conv} A \subseteq \operatorname{conv}(A \cup B)$ and the corresponding inclusion for B. Taking their union proves the claim.

(b) Show strict inclusion can occur.

Let $A = \{(0,0)\}$ and $B = \{(2,0)\}$. The union of their hulls has just two points, whereas the combined hull is the whole segment and includes $(1,0)$.

Exercise 8.3.20: Intersection of hulls can be larger

(a) Apply monotonicity twice.

Since $A \cap B \subseteq A$ and $A \cap B \subseteq B$, its hull is contained in each hull, hence in their intersection.

(b) Give an example with a new intersection point.

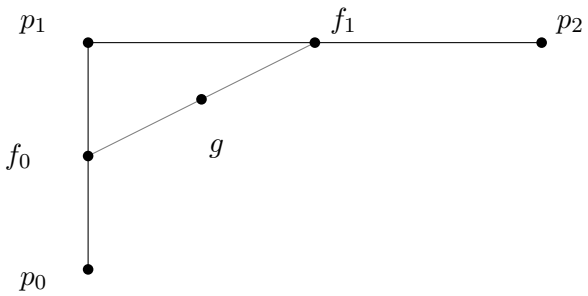
Take $A = \{(-1,0),(1,0)\}$ and $B = \{(0,-1),(0,1)\}$. The sets have empty intersection, so its hull is empty. But their two segments intersect at zero, showing that the claimed inclusion can be strict.

Exercise 8.3.21: The midpoint construction for a quadratic curve

At $t = 1/2$,

$$f_0 = \tfrac{1}{2}(p_0 + p_1), \quad f_1 = \tfrac{1}{2}(p_1 + p_2), \quad g = \tfrac{1}{2}(f_0 + f_1) = \tfrac{1}{4}p_0 + \tfrac{1}{2}p_1 + \tfrac{1}{4}p_2.$$

Thus first mark the midpoints of p_0p_1 and p_1p_2, join them, and mark the midpoint of that new segment. The original diagram below uses the same right-angle configuration as the source, with convenient schematic coordinates $p_0 = (0,0), p_1 = (0,2), p_2 = (4,2)$.

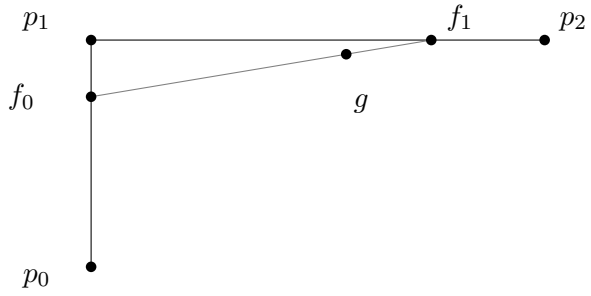


Exercise 8.3.22: A three-quarter construction

Now $f_0 = (1/4)p_0 + (3/4)p_1$ and $f_1 = (1/4)p_1 + (3/4)p_2$. Place each point three quarters of the way along its directed segment. Then place g three quarters of the way from f_0 to f_1:

$$g = \tfrac{1}{16}p_0 + \tfrac{3}{8}p_1 + \tfrac{9}{16}p_2.$$

The same schematic coordinates give $f_0 = (0, 3/2)$, $f_1 = (3, 2)$, and $g = (9/4, 15/8)$, as shown.



Exercise 8.3.23: Expand the quadratic Bézier formula

Substitute both definitions into $g = (1 - t)f_0 + tf_1$ and distribute:

$$g(t) = (1 - t)^2 p_0 + 2t(1 - t)p_1 + t^2 p_2.$$

For $0 \le t \le 1$, the three coefficients are nonnegative and their sum is $(1 - t)^2 + 2t(1 - t) + t^2 = ((1 - t) + t)^2 = 1$. Thus this formula is a convex combination of the control points and lies in their convex hull for the entire parameter interval.

Exercise 8.3.24: The cubic curve also stays inside its controls

Apply the preceding formula to the two quadratic curves and collect terms:

$$h(t) = (1 - t)^3 p_0 + 3t(1 - t)^2 p_1 + 3t^2(1 - t)p_2 + t^3 p_3.$$

The four coefficients are nonnegative on $[0, 1]$ and add to $((1 - t) + t)^3 = 1$ by the binomial theorem. Hence every point of the cubic curve is a convex combination of the four controls and lies in their convex hull. Equivalently, both quadratic points already lie in that convex hull, which is itself convex.